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output (Q). ***AVC = TVC / Q*** where, AVC denotes Average Variable cost, TVC denotes total variable cost and Q denotes quantity of output. For example, When the TVC is ₹ 300 and the quantity produced is 2, the AVC is ₹ 150, (AVC = 300/2 = 150) AVC is shown in table 4.5 and Diagram 4.5. If TVC = $Q^3 - 18Q^2 + 91Q$

AVC = $Q^2 - 18Q + 91$

It is to be noted that

a) AVC declines initially and then increases with the increase of output.

b) AVC declines up to a point and moves upwards steeply, due to the law of returns.

c) AVC curve is a U-shaped curve.

4.4.6 Average Total Cost (ATC) or Average Cost (AC)

It refers to the total cost per unit of output. It can be obtained in two ways.

1. By dividing the firm's total cost (TC) by the quantity of output (Q). ***ATC = TC / Q***. For example, if TC is ₹ 1600 and quantity of output is Q=4, the Average Total Cost is ₹ 400. (ATC = 1600/4 = 400) If ATC is $Q^3 - 18Q^2 + 91Q + 12$, then AC = $Q^2 - 18Q + 91 + {}^{12}/_{Q}$

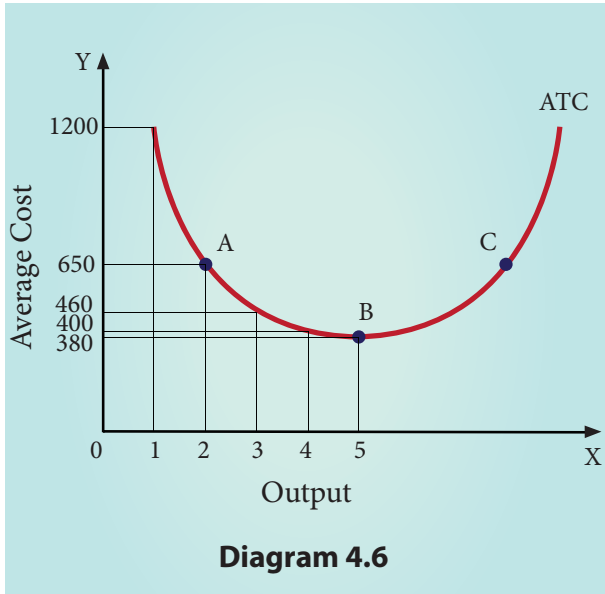


Diagram 4.6

2. ByATC is derived by adding together Average Fixed Cost (AFC) and Average Variable Cost (AVC) at each level of output. ***ATC = AFC + AVC***. For example, when Q= 2, TFC = 1000, TVC=300; AFC=500; AVC=150;ATC=650. ATC or AC is shown in table 4.6 and Diagram 4.6

It should be noted that

Table 4.6 Average Total Costv or Average Cost

Q (in unit)	TFC (in ₹)	TVC (in ₹)	TC (in ₹) TFC +TVC	ATC (TC/Q) (in ₹)	AFC (in ₹)	AVC (in ₹)	ATC (AFC +AVC) (in ₹)
0	1000	0	1000	1000 /0= ∞	0	0	0 + 0 = 0
1	1000	200	1200	1200 /1= 1200	1000	200	1000+200 =1200
2	1000	300	1300	1300 /2= 650	500	150	500 + 150= 650
3	1000	400	1400	1400 /3= 466	333	133	333 + 133= 466
4	1000	600	1600	1600 /4= 400	250	150	250 + 150= 400
5	1000	900	1900	1900 /5= 380	200	180	200 + 180= 380